PRODUCT DEFINITION & ENGINEERING BASELINE

STANDARD VIDEO PLAYER

Software Requirements Specification

A detailed functional, technical, security and operational specification

Document control	Value
Document version	1.0 - Consolidated requirements baseline
Document date	3 November 2026
Product target	Android-first, cross-platform media player
Application stack	Flutter/Dart UI with native C/C++ media and processing components
Distribution assumption	Direct APK distribution initially; manual user-approved installation of updates
Status	Requirements baseline for architecture, UX design and implementation planning
Source	235 user decisions captured during requirements discovery, plus earlier confirmed product constraints

Prepared as an implementation-oriented baseline. This document does not claim that every proposed integration or hardware feature is already implemented; it defines the required behavior, constraints, verification approach and items that must be validated during engineering.

Document control

Field	Description
Purpose	Define the functional and non-functional requirements for the Standard Video Player and its associated cloud, vault, development and administration systems.
Audience	Product owner, UX designer, Flutter engineers, native Android/C++ engineers, backend engineers, QA, security reviewers and release managers.
Baseline authority	Where this document conflicts with older drafts or prototypes, this consolidated baseline takes precedence unless the product owner explicitly approves a change.
Change management	Changes to security, expiry, backup, admin ownership or data-loss behavior require documented impact analysis and approval.
Confidentiality	Contains privileged system design. Secrets, access codes, API credentials, private keys and actual account recovery values are intentionally excluded.

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Executive summary

The Standard Video Player is an Android-first, cross-platform media application designed around a responsive, immersive video player, a dedicated audio experience, a searchable local library, Google Drive media access, utility tools, an encrypted Secret Vault and a layered administration/security environment. The application is intended to provide a strong everyday playback experience without exposing advanced controls to users who do not need them.

The product will use Flutter and Dart for the application and user interface, with native Kotlin/Android integration and C/C++ components for performance-sensitive playback, codec, rendering and media-processing work. The implementation should build on proven media engines and codec libraries rather than attempting to create an entire decoding stack from scratch. Exact engine/library selections remain subject to compatibility, licensing, maintenance and device-benchmark evaluation.

The primary navigation contains Video, Audio, Tools and Settings. The default landing page is the Video Library. The app supports local device media and Google Drive as media sources. Broader cloud-provider support is intended for encrypted backup storage, not as a general replacement for the explicitly limited media-source model. YouTube, YouTube Music, SMB/NAS, DLNA and arbitrary network shares are outside scope.

Standard mode presents essential playback and library functions. Pro Mode is freely available and exposes advanced user-facing controls such as expanded equalizer and color controls, audio/video profiles and advanced enhancement settings. Development Mode is separately gated, time-limited and device-bound; it provides experimental processing and hardware override tools and is activated online through a server-authoritative approval process.

The security model includes a transactional encrypted Secret Vault; encrypted local and cloud backups; device-bound administrator sessions; multi-factor administrator login; tamper-evident audit logs; role-based access; controlled recovery; and emergency lockdown. Backup data is compressed before encryption, each backup receives an independent encryption key, and cloud replicas are verified before the primary copy is removed.

Important engineering boundary

The requirements specify intended behavior, not impossible guarantees. Android cannot guarantee a permanent hardware identity, force a disconnected device to receive a remote command, physically erase every flash-storage remnant, or guarantee that an unsupported cloud provider exposes all required APIs. The implementation must use cryptographic device binding, signed metadata, local expiry enforcement, secure key management and provider capability checks, while clearly reporting residual limitations.

1. Purpose, scope and product vision

1.1 Purpose

This SRS establishes a shared implementation baseline for the Standard Video Player. It defines what the product must do, how its major subsystems interact, which security and operational controls are required, and how completion can be verified. It is intended to reduce ambiguity between product, design, engineering and QA.

1.2 Product vision

Deliver a polished, responsive and immersive media player that handles local and Google Drive media, supports advanced playback and enhancement on capable devices, includes a first-class audio experience, and provides dependable privacy, backup and recovery features. Everyday playback should remain straightforward; advanced and experimental capability should be isolated behind Pro and Development Mode surfaces.

1.3 In-scope capabilities

- Video playback with HW, HW+, SW and hybrid decoder strategies; responsive immersive controls; gestures; resume; bookmarks; playback profiles; HDR handling; frame stepping; A-B repeat; PiP and floating playback.
- Dedicated audio library and player with background playback, queues, playlist import/export, output profiles, equalizer, lyrics, sleep timer and video-as-audio operation.
- Local library scanning and indexing with folder exclusions, hidden-file preference, duplicate detection, search, filters, sorting and user-customizable layouts.
- Google Drive browsing and streaming, optional original-quality offline download, upload/save-back, network controls and time-limited offline-copy governance.
- Utility tools for conversion, audio extraction, frame extraction, subtitle processing, media inspection and batch jobs.
- Pro Mode, Development Mode, graphics capability detection, multi-stage upscaling, deblur, denoise and experimental processing.
- Encrypted Secret Vault with transactional file movement, auto-lock, privacy protections and account-based recovery.
- Local and cloud app backups, restore points, storage offloading, provider integration, administrative exports and recovery.
- Role-based administrator console, protected ownership, 2FA, device management, session controls, audit trails, security monitoring and global lockdown.

1.4 Explicitly out of scope

- YouTube or YouTube Music streaming, downloading, extraction, background playback or account integration.
- SMB, NAS, DLNA or arbitrary network-share playback as media sources.
- Building a complete proprietary codec/decoder engine from first principles.
- Guaranteed recovery of data after loss of all device keys and all valid backups.
- Guaranteed physical secure erase of flash storage or guaranteed remote enforcement while a device remains disconnected.
- Automatic silent installation of APK updates; users must approve and install updates manually.

2. Product goals and design principles

ID	Goal / principle	Product interpretation
G-01	Fast path to playback	Open to Video Library; make scanning, search, resume and play controls obvious.

ID	Goal / principle	Product interpretation
G-02	Progressive disclosure	Keep standard controls clear; put advanced adjustments in Pro; reserve experimental and potentially unstable controls for Development Mode.
G-03	Responsive by design	Adapt layouts to device size, orientation, display cut-outs, multi-window and external displays. Avoid fixed-size layouts.
G-04	Quality-aware processing	Use source-aware, device-aware processing stages. Do not equate a larger output resolution with genuine recovered source detail.
G-05	Privacy by default	Do not transmit diagnostics, analytics or crash reports without the chosen consent.
G-06	Data-loss prevention	Preserve source media by default; use staged file transactions; protect backups with restore points and verified replication.
G-07	Offline-first playback	Local playback and library features must remain useful without a network. Server-dependent activation, cloud browsing and online sync remain network-dependent.
G-08	Honest capability reporting	Report hardware/API/library capabilities accurately; do not advertise unavailable Vulkan, NPU or HDR paths.
G-09	Secure-by-design administration	Admin identity, device binding, authentication, permissions, audit and recovery must be enforced server-side.
G-10	Maintainable extensibility	Use modular interfaces and provider adapters so engines, scalers and cloud providers can be changed without rewriting the application.

3. Stakeholders, users and roles

3.1 User groups

User group	Primary needs	Key capabilities
Standard user	Reliable playback, searchable library, simple controls	Video/audio playback, basic subtitles, standard tools, preferences, local and Drive media.
Pro user	More control without experimental risk	Advanced EQ, color profiles, enhancement tuning, advanced subtitle and player controls.

User group	Primary needs	Key capabilities
Development Mode user	Validate experimental capabilities on approved devices	Experimental scalers, shaders, decoder experiments, hardware overrides, diagnostics and test configurations.
Main Admin / protected owner	Ownership and highest-level governance	Master recovery configuration, administrator management, storage-provider connection, critical security and ownership settings.
Super Admin	Operate the administrative system	Full administrative authority within the protected-owner governance and mutual ownership-transfer rules.
Admin	Operational administration	User requests, offline-download extensions, user management and delegated backup permissions.
Support Admin	Limited support access	View permitted support information and assist with user requests without broad sensitive-data access.

3.2 Role model

ROLE-01 **[Must]** The administrator system shall implement Super Admin, Admin and Support Admin roles with distinct permission sets.

ROLE-02 **[Must]** There shall be no public administrator registration. Admin identities must be pre-approved Google accounts with mandatory 2FA.

ROLE-03 **[Must]** The system shall support a maximum of three Super Admin accounts. The protected owner/Main Admin is one of these accounts.

ROLE-04 **[Must]** The protected-owner designation may transfer to either authorized Super Admin only after mutual approval by the relevant existing Super Admins. The previous protected owner remains a Super Admin unless separately and explicitly changed.

ROLE-05 **[Must]** The Main Admin alone may connect additional backup-storage providers/accounts and change the recovery code. Either Super Admin may independently trigger or deactivate global lockdown under the required reauthentication controls.

ROLE-06 **[Must]** Either Super Admin may grant delegated backup-restoration permission. Backup export and backup restoration shall remain separate permission capabilities.

ROLE-07 **[Must]** Support personnel shall not receive secrets, recovery codes, raw biometrics, private keys or unrestricted access to private media.

ROLE-08 **[Must]** The protected owner shall not be removed or disabled by another Super Admin while protected. Any transfer of the protected-owner designation shall follow the mutual-approval workflow; the former owner remains a Super Admin unless separately changed.

4. Product scope, assumptions and constraints

4.1 Platform strategy

- Android is the first implementation target. The product architecture should remain cross-platform where the selected media, storage and security abstractions permit.

- Flutter/Dart provides the primary UI and application state management. Kotlin provides Android lifecycle, permissions, storage access, notifications, media session, foreground service and platform API integration.
- C/C++ is used for media engine integration, decoder/rendering paths and performance-sensitive processing. A narrow FFI/NDK boundary should keep UI code independent of native implementation details.
- The initial distribution model is direct APK delivery. The app should check for updates and notify the user, but the user manually downloads and approves installation.

4.2 Device and OS constraints

- Android scoped storage and runtime permissions constrain which files can be indexed. The application must request only permissions justified by a feature and use the system file/folder picker where practical.
- Hardware decoder, Vulkan, NPU, HDR output, audio passthrough, external displays and refresh-rate matching are device- and OS-dependent. Capability detection must precede feature exposure.
- Android Keystore may provide hardware-backed key protection on supported devices; the app must detect and report the actual security level rather than assuming hardware backing.
- An Android device identifier is not a universal permanent hardware identity. Device binding must use registered cryptographic keys and risk signals in addition to platform identifiers.

4.3 Network and integration constraints

- Local playback must function offline. Development Mode activation, Google Drive browsing, online subtitle services, updates, cloud backups and cross-device sync require network connectivity.
- Google Sign-In identity, Google Drive media access and backup cloud-provider connections are separate authorization contexts. A Google account login does not automatically authorize Drive file access.
- Cloud backup providers must be integrated using supported official APIs or a vetted connector. Where an API cannot support required upload, download, listing, deletion or verification operations, that provider must be marked unsupported or partially supported rather than accessed through brittle scraping.
- Proton Drive multi-account support is a desired provider capability and must be validated against available official integration options before implementation commitments are made.

4.4 Licensing and distribution constraints

Before selecting mpv, FFmpeg, Media3/ExoPlayer, codec packs, AI models or graphics libraries, engineering must complete license, attribution, redistribution and patent-risk reviews appropriate to the intended distribution model. GPL/LGPL obligations, codec patents and third-party model terms must be assessed by qualified reviewers. This SRS does not grant any third-party license.

5. System architecture and technology strategy

5.1 Logical architecture

Layer	Primary technology direction	Responsibilities
Presentation	Flutter / Dart	Responsive screens, navigation, state, settings, library views, dialogs and user-facing Pro/Dev surfaces.

Layer	Primary technology direction	Responsibilities
Platform integration	Kotlin / Android APIs	Storage and folder access, permissions, background services, media session, PiP, notifications, biometric prompt, device state, external display and APK update handoff.
Media core	C/C++ with selected established engine(s)	Demuxing, decoding, audio/video output, track handling, timestamps, frame stepping, rendering, error recovery and engine abstraction.
Processing pipeline	C/C++ and vetted compute libraries	Deblur, denoise, deblocking, scaling, sharpening, color transforms, frame processing and hardware capability routing.
Local persistence	SQLite plus app-private files	Library index, preferences, playback history, bookmarks, queue, download metadata, vault metadata and job state.
Backend services	Secure API services	Admin identity, Dev Mode activation, device registration, expiry/approval state, recovery, audit records, notifications and cloud job metadata.
Cloud adapters	Provider-specific API adapters	Google Drive media access; backup providers including supported Google Drive, OneDrive, Dropbox and Proton Drive accounts.

5.2 Native boundary and engine strategy

ARCH-01 [Must] The UI layer shall communicate with the native media core through a versioned, typed interface (FFI/platform channel/native plugin) rather than direct scattered native calls.

ARCH-02 [Must] The media core shall expose a stable abstraction for source, decoder, renderer, audio output, tracks, seek, playback state and error events.

ARCH-03 [Must] The implementation shall evaluate established media engines and codec libraries. It shall not reimplement all codec decoding functionality from scratch.

ARCH-04 [Must] Decoder selection shall support HW, HW+, SW and hybrid strategies where supported by the chosen engine/device. The selection policy shall be observable in diagnostics.

ARCH-05 [Must] Native processing modules shall be isolated so that deblur, denoise, scaling, sharpening and color processing can be enabled, disabled, benchmarked and replaced independently.

ARCH-06 [Must] GPU/NPU/CPU paths shall be capability-probed and validated. Unsupported paths must fall back safely without blocking ordinary playback.

5.3 Backend authority

The backend is authoritative for administrator roles, Dev Mode activation/expiry/revocation, protected-owner state, offline-download extension approvals, admin recovery records, audit events and cloud backup metadata. A mobile client must not be able to grant itself Dev Mode, extend an expiry, alter administrator roles or erase audit history by editing local data.

5.4 Modularity

- Media engine adapter
- Decoder policy and fallback controller
- Video processing graph
- Audio processing graph
- Local library scanner/indexer
- Google Drive media adapter
- Backup provider adapters
- Vault encryption and transaction manager
- Admin/security client
- Download and conversion job manager

6. Information architecture and navigation

6.1 Primary navigation

Area	Purpose	Key content
Video	Default landing destination	Video library, search/filter/sort, layouts, favorites, history/resume and video playback.
Audio	Dedicated audio destination	Audio library, playlists, albums/artwork, queue, lyrics, audio playback and output settings.
Tools	Utility workspace	Converters, extraction, subtitle utilities, frame capture, media details and batch jobs.
Settings	Preferences and management	Appearance, language, scanning, downloads, playback defaults, privacy, notifications, backup and connected accounts.

A persistent coin icon on major pages opens the Earn area. The current Earn destination is a Coming Soon placeholder; no earning, rewards or payment workflow is included in this release baseline.

6.2 Library layouts

UX-01 **[Must]** The Video and Audio libraries shall be separate, with independent layout, sort, filter and view preferences.

UX-02 **[Must]** The library shall support user-customizable layouts, including grid, list and compact grid as baseline arrangements.

UX-03 **[Must]** The library shall support global search and section-specific search.

UX-04 **[Must]** The library shall provide resolution filters for all detected resolutions, including 720p, 1080p, 2K, 4K, 8K and other detected values.

UX-05 **[Must]** Sort/group options shall include name, date, size, duration, resolution, folders, format and recently played.

UX-06 **[Must]** User preferences shall persist across restarts and remain independently configurable for Video and Audio.

UX-07 **[Must]** Customizable library layouts shall allow the user to adjust layout mode and supported presentation properties such as thumbnail density/size, columns, visible metadata and spacing, subject to responsive layout constraints.

6.3 Visual and responsive design

- All screens shall adapt to available width/height, density, orientation, window size and display cut-outs.
- No core page shall rely on a fixed absolute-size canvas that clips on smaller or larger devices.
- The app shall support Light, Dark and System themes. Additional downloadable or bundled skins may be introduced in later versions and must be clearly labeled Coming Soon until implemented.
- The default opening destination is Video Library.
- Language shall follow device language by default, with manual language selection.
- App-specific accessibility controls shall complement Android accessibility settings.

7. Functional requirements

7.1 Startup, permissions and settings

APP-01 **[Must]** On first launch, onboarding shall explain the permissions needed for the initial library/player experience and request those necessary permissions. Feature-specific permissions that cannot or should not be requested at startup shall be requested when the related feature is first used, following Android policy.

APP-02 **[Must]** The app shall provide a settings switch for background audio playback.

APP-03 **[Must]** The app shall provide individual notification preferences and a master notification switch.

APP-04 **[Must]** The app shall ask for consent during first installation/onboarding before collecting analytics or usage data. If consent is denied, collection must remain disabled. Consent must be changeable later in Settings.

APP-05 **[Must]** Crash reports and diagnostic reports shall require explicit per-report consent unless the user separately enables a future-consent preference. A crash dialog shall clearly state that the app crashed and offer an optional report submission.

APP-06 **[Must]** App data clearing shall offer a backup opportunity and then allow the user to proceed only after explicit confirmation.

APP-07 **[Must]** App launch lock shall be optional and support PIN and device biometric authentication.

APP-08 **[Must]** The app shall ask for explicit confirmation before deleting any media file. Duplicate cleanup and repair utilities shall not delete original files without confirmation.

7.2 Local media sources and scanning

LIB-01 **[Must]** The app shall scan all accessible media locations by default, subject to Android permission and scoped-storage rules.

LIB-02 **[Must]** Users shall be able to filter included folders and exclude folders and file types.

LIB-03 **[Must]** Hidden files and folders shall be excluded by default, with an opt-in setting to include them.

LIB-04 **[Must]** The scanner shall detect newly added media automatically where OS facilities permit and shall also run scheduled full rescans.

LIB-05 **[Must]** The scanner shall identify likely duplicate files and offer to hide duplicate library entries or remove an entry. It shall not silently delete the underlying media file.

LIB-06 **[Must]** The library shall store media references and technical metadata in a local database; it shall not unnecessarily copy source media into the app data directory.

LIB-07 **[Must]** The scanner shall tolerate inaccessible, transient, unsupported or corrupted files and continue scanning other items.

LIB-08 **[Must]** Users shall be able to rescan folders and see scan progress and recoverable errors.

7.3 Video playback

VID-01 **[Must]** The player shall provide an immersive full-screen mode that uses the available display area and adapts responsively to device dimensions.

VID-02 **[Must]** The video title shall remain hidden during immersive playback unless the user explicitly opens information or playback controls.

VID-03 **[Must]** A single tap shall reveal standard controls; controls shall auto-hide after a configurable short delay.

VID-04 **[Must]** Standard controls shall include Play/Pause, Previous Video and Next Video. Previous/Next shall switch media items and shall not act as 10-second seek controls.

VID-05 **[Must]** Double-tap on the left or right playback area shall seek backward or forward by 10 seconds respectively.

VID-06 **[Must]** The player shall support HW, HW+, SW and hybrid decoder modes where supported, with automatic selection and fallback after a decoder failure.

VID-07 **[Must]** On a hardware-decoder failure, the player shall automatically attempt a compatible alternative decoder path, including software or hybrid mode where appropriate.

VID-08 **[Must]** For damaged frames, the player shall attempt recovery; if recovery fails, it shall skip the damaged frame and continue where safe.

VID-09 **[Must]** On a playback failure, the player shall retry through another compatible playback method before surfacing an unrecoverable error.

VID-10 **[Must]** On unexpected crash, the app shall clearly disclose the crash and offer optional report submission. The report must not be sent without consent.

VID-11 **[Must]** The player shall support the widest stable playback-speed range exposed by the engine and device. The UI shall validate the actual supported range rather than promising a fixed maximum.

VID-12 **[Must]** The player shall support dedicated previous-frame and next-frame controls where frame stepping is supported by the active engine.

VID-13 **[Must]** The player shall support A-B loop markers and repeat playback between the selected timestamps.

VID-14 **[Must]** The user shall be able to choose resume behavior in Settings. The app shall not repeatedly ask on every playback start.

VID-15 **[Must]** Playback history shall be retained indefinitely until the user clears it. Resume positions shall be saved with sufficient precision for practical restoration.

VID-16 **[Must]** The player shall support global or per-video brightness and volume preferences, configured in Settings rather than prompted every time.

VID-17 **[Must]** The player shall follow device orientation by default and provide a temporary rotation control.

VID-18 **[Must]** The player shall support Android Picture-in-Picture and a separate floating player window where permitted by OS/device policy.

VID-19 **[Must]** The player shall match display refresh rate to video frame rate where supported, with safe fallback if the OS denies or does not support the request.

VID-20 **[Must]** The player shall use best supported HDR output when the display path supports HDR; otherwise it shall tone-map HDR content to SDR.

VID-21 **[Must]** The player shall expose expandable panels for subtitles, audio tracks, speed, aspect ratio, rendering and enhancement options. Advanced controls shall not overwhelm the default screen.

VID-22 **[Must]** The player shall support Eco, Balanced and Performance playback modes. Automatic thermal/battery adaptation shall be available with a manual override option.

VID-23 **[Must]** Playback profiles shall be selected automatically based on hardware and video characteristics, while allowing users to create and select custom profiles.

VID-24 **[Must]** The player shall support compatible external displays and external audio output.

VID-25 **[Must]** The Video Library shall provide a prominent Resume entry for the last interrupted item when a valid resume position exists, using the configured resume behavior and exact saved timestamp.

7.4 Gestures and controls

CTL-01 **[Must]** The default gesture map shall assign left-side vertical gestures to brightness, right-side vertical gestures to volume and horizontal gestures to seeking.

CTL-02 **[Must]** Standard controls shall remain usable with touch, accessibility services and supported remote/headset controls.

CTL-03 **[Must]** Advanced customization of player buttons, gesture mappings, button sizes, visibility and auto-hide timing shall be available in Pro Mode.

CTL-04 **[Must]** The center Previous/Next controls shall preserve their media-navigation semantics even when users customize layout.

7.5 Audio player

AUD-01 **[Must]** The app shall provide a first-class Audio library separate from Video.

AUD-02 **[Must]** Audio playback shall support background playback and screen-off playback. Users shall control this behavior in Settings.

AUD-03 **[Must]** The audio player shall expose configurable headset/Bluetooth media controls, including play/pause, next and previous where the device supports them.

AUD-04 **[Must]** The audio player shall support automatic folder progression and a user-managed playback queue.

AUD-05 **[Must]** The application shall support playlists and M3U/PLS import/export, plus app-level backup and restoration of playlist structures.

AUD-06 **[Must]** The app shall support shuffle, repeat, album artwork and queue management.

AUD-07 **[Must]** Standard mode shall expose normal equalizer controls. Pro Mode shall expose advanced equalizer controls and richer profile customization.

AUD-08 **[Must]** Standard users shall be able to select available audio presets. Pro users shall be able to tune and save custom audio profiles.

AUD-09 **[Must]** The app shall provide clipping protection and a limiter for boosted playback. Volume boost shall support the requested up-to-100-percent boost range where the output path safely permits it, with explicit warning that digital gain cannot guarantee a louder physical output on every device.

AUD-10 **[Must]** The app shall support local and online lyrics, with synchronized lyrics where the provider and track metadata permit.

AUD-11 **[Must]** The app shall provide a sleep timer based on elapsed time and/or end of current track.

AUD-12 **[Must]** The app shall detect phone speakers, wired headphones and Bluetooth output where Android exposes the route, then apply suitable automatic profiles. Users may override the automatic profile.

AUD-13 **[Must]** Audio passthrough shall be enabled automatically only when the device, output route, codec and application path support it; otherwise audio shall use a compatible decoded output path.

AUD-14 **[Must]** The app shall support Video-as-Audio playback by playing the audio track without rendering the video.

AUD-15 **[Must]** Bit-perfect or lossless output shall only be claimed when the end-to-end output path actually preserves the source. Bluetooth and device audio processing may prevent bit-perfect delivery.

7.6 Subtitle functionality

SUB-01 [Must] The player shall support commonly used subtitle formats through validated libraries/codecs, including SRT, ASS, SSA, SUB, VTT and image-based formats where supported.

SUB-02 [Must] The player shall automatically locate matching local subtitle files where practical and allow manual subtitle selection.

SUB-03 [Must] The app shall support online subtitle search/download through supported providers and shall keep online access user-initiated or clearly configurable.

SUB-04 [Must] Subtitle preferences shall default to device language, with the ability to change preferred languages and priority in Settings.

SUB-05 [Must] When multiple subtitle tracks are available, the player shall present a selection interface rather than silently selecting an arbitrary track.

SUB-06 [Must] Subtitle synchronization shall include manual timing offset and automatic synchronization assistance where feasible.

SUB-07 [Must] Standard subtitle appearance controls shall include font, size, color, outline, background and position; Pro Mode may provide advanced styling options.

SUB-08 [Must] Subtitle tools shall support synchronization, conversion, merge, split and editing operations where the format permits.

7.7 Search, favorites and history

SRCH-01 [Must] The app shall provide global search across media, folders, playlists and supported tool items, as well as separate Video and Audio search.

SRCH-02 [Must] Video and Audio favorites shall be separate collections.

SRCH-03 [Must] Playback history shall be persistent until manually cleared and shall include resume references and relevant metadata.

SRCH-04 [Must] Bookmarks shall save playback timestamps. Bookmark export/import is not required in the current baseline unless added to a later version.

7.8 Tools and media conversion

TOOL-01 [Must] The Tools section shall provide video format conversion and device-ready conversion presets.

TOOL-02 [Must] Before conversion, the user shall select or confirm the target quality/resolution, such as 360p source to 1080p output. The app shall not silently claim that upscaling restores native source detail.

TOOL-03 [Must] Upscaling conversion shall use a staged processing graph where appropriate. Candidate stages include decode/normalization, artifact reduction, deblur, scaling, sharpening/color correction and final encoding. Each stage shall be independently testable and tunable.

TOOL-04 [Must] The default conversion quality policy shall choose appropriate bitrate, codec and other parameters for the selected target, while explaining output expectations. Advanced manual encoder controls may be added in Pro/Dev as appropriate.

TOOL-05 [Must] The user shall choose whether to preserve the original, create a separate output or replace the original after explicit confirmation. The preference may be remembered and changed in Settings.

TOOL-06 [Must] The conversion manager shall support multiple queued jobs and parallel conversion only when device capability, thermal state and memory allow it.

TOOL-07 [Must] Conversion queue controls shall include pause, resume, cancel, reorder and retry, with per-file progress and error reporting.

TOOL-08 **[Must]** Audio extraction shall preserve the original audio stream without re-encoding where technically possible; otherwise it shall offer supported formats such as MP3, AAC, WAV, FLAC and other library-supported formats.

TOOL-09 **[Must]** Frame tools shall support single-frame capture, multi-frame extraction and contact-sheet/thumbnail-sheet generation.

TOOL-10 **[Must]** Media information shall show a basic view in Standard mode. Pro Mode may expose detailed technical diagnostics such as codec, bitrate, resolution, frame rate, tracks, HDR metadata and decoder path.

TOOL-11 **[Must]** Batch operations shall be supported across compatible selected media files.

TOOL-12 **[Must]** Repair tools shall diagnose damaged media and, where a safe repair is possible, write a separate repaired output. The source shall remain untouched unless the user explicitly chooses otherwise.

TOOL-13 **[Must]** The tool workspace shall support job cancellation and safe cleanup of partial outputs.

8. Development Mode, Pro Mode and experimental capabilities

8.1 Standard mode

Standard mode is the everyday experience. It includes essential playback, ordinary equalizer controls, standard selectable audio presets, core subtitle controls, library features and safe automatic enhancement when the device supports it. It should not expose experimental hardware overrides or low-level scaler tuning.

8.2 Pro Mode

PRO-01 **[Must]** Pro Mode shall be freely unlocked and shall not require a code, passphrase or separate purchase gate in the current product baseline.

PRO-02 **[Must]** Pro Mode shall expose advanced color controls, at least ten color profiles, including an AMOLED-oriented black profile, and user adjustment where supported.

PRO-03 **[Must]** Pro Mode shall expose at least ten audio profiles/presets, such as Jazz, Pop and other common listening profiles, plus custom tuning.

PRO-04 **[Must]** Pro Mode shall expose advanced equalizer controls, aspect-ratio controls, playback speed controls, expanded subtitle styling and advanced playback-control customization.

PRO-05 **[Must]** Pro Mode shall expose advanced enhancement intensity settings, separate deblur and upscaling controls, and clear before/after preview tools.

PRO-06 **[Must]** On entry, Pro Mode shall display detected graphics capabilities (for example Vulkan support or unavailable status) and related hardware capabilities before presenting advanced controls.

PRO-07 **[Must]** Pro Mode shall expose the requested volume boost up to 100% where safe, with a limiter and clipping protection. The app must not promise that this doubles the real acoustic output on every device.

PRO-08 **[Must]** Pro Mode shall keep advanced settings organized in expandable sections rather than placing all controls on the primary playback screen.

8.3 Automatic enhancement and upscaling

ENH-01 **[Must]** When supported hardware can handle it, automatic upscaling shall be enabled according to a hardware/source-aware recommendation.

ENH-02 **[Must]** The automatic enhancement pipeline shall choose stages based on source quality, motion/content characteristics, device capability and thermal/battery state.

ENH-03 **[Must]** Deblur shall be a separately implemented processing subsystem with its own strength, thresholds, artifact checks, enable/disable state and test coverage.

ENH-04 **[Must]** Noise and compression-artifact reduction shall be a separate subsystem with configurable strength and safeguards against loss of legitimate texture.

ENH-05 **[Must]** Upscaling shall be a separate subsystem with selectable scaler/model paths and output validation.

ENH-06 **[Must]** Sharpening and color correction shall be optional stages and shall not be forced when they introduce ringing, halos or color shifts.

ENH-07 **[Must]** Pro Mode shall expose safe user controls; Development Mode shall expose experimental stage overrides and low-level hardware choices.

ENH-08 **[Must]** Before/after preview shall support both side-by-side comparison and an adjustable comparison slider.

ENH-09 **[Must]** The system shall adapt processing intensity when battery or thermal conditions require it. Users may override adaptation where safe, but the app must retain protective limits against device instability.

ENH-10 **[Must]** The system shall report when a target output such as 1080p is achieved by upscaling a lower-resolution source. It shall not represent the result as equivalent to native 1080p detail.

8.4 Development Mode access and lifecycle

DEV-01 **[Must]** Development Mode activation shall require a specific authorization code and a live online validation against the authoritative backend.

DEV-02 **[Must]** Development Mode access shall be time-limited, device-bound and revocable. The backend shall record activation, start time, expiry, usage events, opening/closing, failed attempts and revocation.

DEV-03 **[Must]** Development Mode shall automatically grant Experimental Access; no separate experimental-access code is required.

DEV-04 **[Must]** Development Mode shall include custom shaders, scalers, experimental processing, hardware/decoder experiments, diagnostics and experimental configuration tools.

DEV-05 **[Must]** Manual overrides to hardware selection, scaler selection and experimental processing shall require Development Mode. Automatic hardware-recommended upscaling remains available outside Development Mode.

DEV-06 **[Must]** When Development Mode expires or is revoked, experimental overrides shall be reset and recommended hardware settings restored. Standard and Pro features shall remain available.

DEV-07 **[Must]** The service shall support up to three linked devices per user account. Each additional eligible device shall receive an independent Development Mode approval and activation period.

DEV-08 **[Must]** The second and third device activation requests shall be submitted to the designated project contact for manual review, with an expected 24-48 hour review window as an operational target, not a guaranteed SLA.

DEV-09 **[Must]** A fourth linked device request shall show existing linked devices and require the user to choose one to replace, confirm the replacement, reauthenticate and revoke the replaced device. The event shall be audited and the owner notified.

DEV-10 **[Must]** Development Mode activation shall not be transferred by application backup or restored settings. A backup must never unlock Development Mode.

DEV-11 **[Must]** One day before Development Mode expiry, the app shall prompt the user to export compatible player settings. On import, non-development settings may be restored when Dev Mode is inactive; development-only settings shall be ignored unless the target device has valid Dev Mode authorization.

9. Google Drive media integration and offline downloads

9.1 Source boundaries and sign-in

DRV-01 [Must] The app shall support local device storage and Google Drive as the only approved media-source categories in the current baseline.

DRV-02 [Must] Google Sign-In for app identity/device linking shall be separate from Google Drive authorization and scopes.

DRV-03 [Must] The app shall allow the user to connect a Google Drive account, browse folders, identify supported media and stream compatible media without requiring a full local download first.

DRV-04 [Must] The app shall allow optional download of an original-quality Drive file and upload/save-back of supported video files to Drive.

DRV-05 [Must] The app shall not integrate YouTube or YouTube Music. YouTube extraction, downloading, streaming and background playback are explicitly excluded.

9.2 Download manager

DLD-01 [Must] The download manager shall support multiple queued downloads, pause, resume, retry, progress, priority/reordering, cancellation and removal from queue.

DLD-02 [Must] The user shall choose, per download, between private app storage hidden from ordinary gallery/file-manager views and a user-selected folder accessible through the file manager, subject to Android storage permissions.

DLD-03 [Must] Offline downloads shall preserve the original file bytes/quality. The system shall not compress, transcode or reduce quality as part of offline download.

DLD-04 [Must] Concurrent download count shall adapt to network, device capability, memory, battery and thermal state, with a manual override subject to safe limits.

DLD-05 [Must] Network controls shall include Wi-Fi-only, mobile-data allowance, ask-before-mobile-data, roaming restriction and data-use warnings/limits.

DLD-06 [Must] Automatic download rules shall be optional and disabled by default. The user may define selected folders, media types, new-file rules, network constraints and storage thresholds.

DLD-07 [Must] Automatic-download storage shall have a user-configured limit with suggested presets such as 5 GB, 10 GB, 20 GB or custom. At the limit, the app shall ask the user whether to expand the limit, delete selected offline copies or cancel; it shall not silently delete or expand storage.

DLD-08 [Must] Background Google Drive download behavior shall be configurable in Settings, including whether downloads continue while the app is in the background and whether persistent notifications are shown.

9.3 Expiry and extension lifecycle

EXP-01 [Must] Each offline Drive copy shall have an initial validity period of 30 days, with a five-day grace period after expiry.

EXP-02 [Must] The app shall issue daily expiry warnings through both in-app notices and Android system notifications, including specific remaining-time warnings such as two days remaining.

EXP-03 [Must] The user may request an extension by contacting the designated project email within the five-day grace period. The countdown shall continue while the request is pending.

EXP-04 [Must] An Admin approval shall apply to all offline downloads owned by that user, not an individual file.

EXP-05 [Must] An approved extension shall add a period selected by an authorized Admin between 1 and 30 days.

EXP-06 **[Must]** Only one extension request may be pending per user. A second request shall be rejected or disabled until the first is resolved.

EXP-07 **[Must]** If a request is rejected, the user shall be notified immediately. The user may submit another request while still within the original five-day grace period.

EXP-08 **[Must]** If no extension is approved by the end of the grace period, the app shall delete the affected offline copy when the device is able to enforce the local expiry policy. The original Drive file shall remain untouched.

EXP-09 **[Must]** Monthly continuation requires another approval request. The system shall define the next due/expiry timestamp explicitly after each extension to avoid ambiguous overlapping monthly and 30-day cycles.

EXP-10 **[Must]** The backend shall be authoritative for extension approval and expiry metadata. Signed records shall be cached locally so that the client can enforce expiry while offline.

EXP-11 **[Must]** The app shall be honest that remote deletion cannot be guaranteed while a device is powered off or disconnected. Local deletion shall occur at the next enforceable opportunity; server state shall prevent renewed access when connectivity returns.

EXP-12 **[Must]** The system shall sync permitted download names/references, stable Drive IDs, metadata, expiry dates and approval history by account. It shall not upload the actual downloaded media files to this administrative metadata service.

9.4 Drive caching and sync

DRV-06 **[Must]** Drive caching shall support Minimal, Balanced and Aggressive levels, with Balanced as the default.

DRV-07 **[Must]** Offline changes shall be saved locally and synchronized when internet connectivity returns.

DRV-08 **[Must]** Sync shall be hybrid: important changes may sync immediately; ordinary changes may sync periodically in the background.

DRV-09 **[Must]** Conflict resolution shall compare timestamps and use the latest valid update where unambiguous. Ambiguous conflicts shall be shown to the user for manual resolution.

DRV-10 **[Must]** Offline-library synchronization shall not upload source media unless the user explicitly requests such an operation.

10. Secret Vault and privacy controls

10.1 Vault protection model

VLT-01 **[Must]** The Secret Vault shall encrypt media and associated metadata, including names and thumbnails, so that plaintext metadata is not exposed through ordinary library indexing.

VLT-02 **[Must]** The vault shall use authenticated encryption such as AES-256-GCM with unique nonces and managed keys. Key material shall be protected with Android Keystore where available, preferably hardware-backed.

VLT-03 **[Must]** Vault media shall not be exposed through Android MediaStore or ordinary gallery indexing.

VLT-04 **[Must]** Vault playback shall decrypt data for playback through a controlled stream or temporary buffer. It shall not create persistent plaintext extraction files unless a separately authorized export operation is added.

VLT-05 **[Must]** The Vault shall support optional app PIN and biometric access controls, plus configurable auto-lock on backgrounding and after inactivity.

VLT-06 **[Must]** Screenshots and screen recording shall be blocked while Vault content is visible, subject to Android platform capabilities.

VLT-07 **[Must]** Sensitive previews shall be hidden from the Android Recent Apps screen.

10.2 Transactional move into Vault

1. Obtain the relevant file/folder access authorization and verify the source can be read.
2. Create a unique staging record and encrypt into private app storage. Do not delete or overwrite the source at this stage.
3. Calculate and verify integrity/authentication metadata for the encrypted result; confirm all expected bytes and metadata are present.
4. Commit the encrypted vault object and metadata atomically, or mark the transaction incomplete if the commit fails.
5. Only after a successful commit and user-authorized move operation may the app delete the original file. If any step fails, roll back staging and preserve the original.
6. On crash recovery, detect incomplete transactions and either complete verification or restore a safe pre-commit state. Do not create duplicate canonical vault entries.

Flash-storage limitation

Application-level deletion can remove directory references and overwrite accessible temporary data, but flash translation layers, snapshots and wear-leveling may retain physical remnants. The app must never promise forensic-grade physical secure erase.

10.3 Vault key recovery

VLT-08 **[Must]** Vault recovery shall be based on the user Google account without requiring an additional user-created recovery password, as selected for the Vault feature.

VLT-09 **[Must]** The backend may maintain server-side encrypted key escrow. It shall never store or expose plaintext vault keys to administrators or support dashboards.

VLT-10 **[Must]** When restoring to another device, the recovered key shall be re-protected under the new device key and account context before use.

VLT-11 **[Must]** Vault recovery shall remain distinct from administrator recovery-code recovery and from app-settings backup password recovery.

VLT-12 **[Must]** Recovery shall be audited and should notify the account owner. Account compromise is a residual risk and must be mitigated with Google account security, device checks and recovery alerts.

11. Backup, restore and cloud-storage management

11.1 Backup scope

BKP-01 **[Must]** The application shall support manual and automatic backups.

BKP-02 **[Must]** The scheduled backup cadence shall include a daily incremental backup for current data and a weekly full backup. Critical metadata shall be protected by the daily schedule, and important changes, imports, migrations and recovery operations shall trigger additional restore points.

BKP-03 **[Must]** Incremental backups shall capture changed data relative to a verified baseline/chain. The system shall track dependencies, validate each incremental chain and preserve the required full-backup base before any authorized deletion or offload operation.

BKP-04 **[Must]** App settings backups shall include settings, custom profiles, playlists/favorites metadata, titles and references, playback history/resume positions, bookmarks, subtitle preferences, folder names and indexing configuration.

BKP-05 **[Must]** Ordinary app-settings backups shall not include actual local media files or Google Drive credentials/tokens. Media files shall only be included by a separate explicit user-authorized media backup operation if such a feature is later approved.

BKP-06 **[Must]** Administrator backups shall be comprehensive and include administrator accounts/roles, user records, request and approval history, security logs, device history, backup configuration and administrative settings, while excluding plaintext secrets, private keys and active session tokens.

BKP-07 **[Must]** Cross-device app backup restoration shall require the same Google account and the user-created backup password where applicable. Vault key recovery remains a separate account-based process.

11.2 Compression and encryption

BKP-08 **[Must]** Backup archives shall be compressed before encryption.

BKP-09 **[Must]** Each backup shall receive a unique randomly generated encryption key. The archive shall use authenticated encryption such as AES-256-GCM.

BKP-10 **[Must]** Backup-key recovery shall support both the designated recovery-code pathway and the backup-password pathway selected for the backup system. These credentials shall not be used directly as raw AES keys.

BKP-11 **[Must]** Encryption keys shall be wrapped/derived using approved key-management mechanisms. The system shall use strong password-based KDF parameters for password-derived protection and shall not embed secrets in the APK.

BKP-12 **[Must]** Backups shall be encrypted locally before upload. Data in transit shall use TLS 1.3 where supported; stored backups shall receive provider/server at-rest protection in addition to archive encryption.

BKP-13 **[Must]** Backup files shall use randomized filenames in cloud storage. Local interfaces may show readable metadata without exposing secrets.

BKP-14 **[Must]** No plaintext password, recovery code, master access code, raw biometric data, private key or active session token shall appear in a backup, filename, dashboard or log.

11.3 Storage quota and retention

BKP-15 **[Must]** The primary app/server backup storage target shall be capped at 5 GB by default. Connected cloud storage shall provide additional capacity outside this primary quota.

BKP-16 **[Must]** When primary storage approaches 70% usage, the system shall begin evaluating oldest-backup offloading according to the configured policy.

BKP-17 **[Must]** The system shall offload the minimum number of oldest eligible backups needed to release required primary storage capacity.

BKP-18 **[Must]** After a cloud replica has passed upload, encryption, integrity and successful retrieval verification, the primary-storage archive may be removed while its metadata, unique ID, verification record and restore-point entry remain.

BKP-19 **[Must]** Cloud-offloaded backups shall be retained indefinitely until explicitly deleted by an authorized administrator.

BKP-20 **[Must]** When the primary 5 GB storage limit is reached and space cannot be safely freed, the system shall ask whether to replace eligible older backups or connect additional cloud storage. It shall never silently delete or overwrite a restore point.

BKP-21 **[Must]** When an offload account is disconnected, full or unavailable, the system shall pause offloading and ask the Main Admin whether to use another connected account or stop. It shall not automatically switch to another account without that decision.

BKP-22 **[Must]** Only one secondary cloud account shall receive a specific backup replica at a time under the current baseline. Multiple accounts may be connected for capacity planning and sequential offload.

BKP-23 [Must] Manual movement of backups between connected accounts is not required. Offloading shall follow the automatic policy.

11.4 Provider support

BKP-24 [Must] The backup subsystem shall use extensible provider adapters for supported cloud storage services.

BKP-25 [Must] Desired providers include Google Drive, OneDrive, Dropbox and Proton Drive, including multiple accounts for a supported provider.

BKP-26 [Must] The Main Admin alone shall connect additional backup-storage accounts/providers.

BKP-27 [Must] Before a provider is enabled, the adapter shall test required upload, download, listing, deletion and retrieval operations using temporary test data where the provider supports them.

BKP-28 [Must] The storage dashboard shall show connected provider accounts, capacity when available, used/free space, backup allocations, connection health and last successful verification.

BKP-28A [Must] Connected cloud accounts shall receive a scheduled daily connection and capacity/health check, with provider-specific limits clearly reported.

11.5 Restore points and integrity

BKP-29 [Must] Restore points shall be created before major administrative changes, imports, migrations, backup operations and recovery operations.

BKP-30 [Must] The system shall retain as many restore points as the storage policy allows. If capacity requires replacement, the Main Admin shall be asked before older restore points are removed.

BKP-31 [Must] Every backup shall receive a unique ID and a tamper-evident verification record.

BKP-32 [Must] Integrity shall be verified after creation, during periodic checks and before restoration.

BKP-33 [Must] Weekly isolated restoration tests shall verify that backups can be decrypted and restored in a controlled environment without overwriting production data.

BKP-34 [Must] If a backup is corrupted, the system shall attempt the previous valid backup and notify both Super Admins. The corrupted object should be retained for investigation if safe and storage permits.

BKP-35 [Must] Backup and restore operations shall support pause/resume where technically safe.

BKP-36 [Must] Failed backup jobs shall retry up to three times with increasing delays. If all retries fail, the process shall stop and create a request for Main Admin approval before further retries or manual intervention.

BKP-37 [Must] Backup health reports shall be generated weekly and monthly.

BKP-38 [Must] Backup creation and restoration shall notify both Super Admins through email, admin dashboard and Android push notification.

11.6 Restore permissions and behavior

BKP-39 [Must] Backup export and backup restoration shall be separate permissions.

BKP-40 [Must] Any administrator granted backup-restoration permission may perform authorized restoration. Either Super Admin may grant that permission.

BKP-41 [Must] The restore interface shall support both full-system restoration and selective restoration by data category.

BKP-42 [Must] Before restoration, the interface shall preview restore point date, contents, versions and detected conflicts.

BKP-43 [Must] The system shall merge restored data while preserving newer valid records where possible. Conflicts that cannot be resolved reliably shall be surfaced for administrator review.

BKP-44 **[Must]** Before every restore, the system shall automatically create a backup/restore point of the current state.

BKP-45 **[Must]** Full-system restoration shall require maintenance mode and approval from another Super Admin. Selective restoration shall follow delegated permission without mandatory second-admin approval unless the selected category is marked critical.

BKP-46 **[Must]** The system shall support restoration on a new server or fresh installation after identity, authorization and key-recovery verification.

BKP-47 **[Must]** Restoration shall be transactional. If the operation fails, the system shall roll back to the pre-restore state or preserve a recoverable snapshot.

11.7 Removed administrator records

BKP-48 **[Must]** When an administrator is removed, the system shall package their relevant administrative records into an encrypted archive (such as an encrypted ZIP) and make it securely available to all configured Main Admins and Super Admins.

BKP-49 **[Must]** The archive shall be distributed through authenticated, access-controlled download links or a secure admin portal; raw sensitive data shall not be attached unencrypted to ordinary email.

BKP-50 **[Must]** The archive shall preserve audit history and record who accessed or downloaded it. Its retention and deletion shall follow the administrative retention policy.

BKP-51 **[Must]** Only Super Admins may manually delete restore points.

BKP-52 **[Must]** Deleting a backup or restore point shall require fresh password/2FA verification and explicit confirmation. The system shall identify dependent incremental chains before deletion and preserve any required recovery base.

12. Administrator console and security governance

12.1 Login and session policy

ADM-01 **[Must]** Administrator login shall use pre-approved Google accounts and mandatory 2FA. There shall be no public sign-up.

ADM-02 **[Must]** A new administrator device shall require Google login and 2FA only; approval from another Super Admin or an existing device is not required for the ordinary new-device login path.

ADM-03 **[Must]** A new-device login notice shall be sent only to the Super Admin who signed in. This is an explicit exception to broader security-event notifications.

ADM-04 **[Must]** Each Super Admin may have a maximum of two active administrator devices/sessions. A third login shall require the Super Admin to choose an existing session to terminate.

ADM-05 **[Must]** Admin sessions shall expire after 30 minutes of inactivity.

ADM-06 **[Must]** Critical operations shall require fresh authentication and 2FA even within an otherwise active session. Examples include ownership transfer, master/recovery code changes and global-lockdown deactivation.

ADM-07 **[Must]** The device list shall show detected device name/model, custom name, login date, last activity and active/revoked state. A Super Admin may rename their own device.

ADM-08 **[Must]** A Super Admin shall be able to revoke an individual device/session. Revocation shall immediately terminate active access and revoke associated tokens.

ADM-09 **[Must]** After revocation, re-registration of that device shall require a fresh Google login and 2FA.

ADM-10 **[Must]** Failed login attempts shall trigger progressive lockout periods and immediate alerts to the affected Super Admin. Attempts and signals shall be recorded without logging credentials or OTP values.

12.2 Account recovery and recovery code

REC-01 [Must] If a Super Admin loses access, the system shall first evaluate device ownership using registered device cryptographic identity, device-bound keys, prior registration records and risk signals.

REC-02 [Must] Device verification shall not rely solely on Android ID, model name, IP address or other spoofable fingerprints. Hardware-backed cryptographic proof shall be used when available.

REC-03 [Must] If device verification is unavailable or fails because the device/account is lost, the designated recovery/reset code shall enable the controlled recovery path to reset access, add/replace administrator email identities and import an encrypted administrator backup.

REC-04 [Must] The recovery code shall remain valid after successful recovery until the Main Admin changes it.

REC-05 [Must] Changing the recovery code shall require the existing recovery code, Google login and 2FA. The Main Admin alone may change it.

REC-06 [Must] The recovery code shall be stored as a secure verifier or protected key-wrapping input, never in plaintext logs, dashboards, APK resources or ordinary database records.

REC-07 [Must] The bootstrap recovery value supplied by the product owner shall be provisioned through a secure deployment/setup workflow. The value shall not be printed in this SRS. Because a human-selected word is low entropy, production deployment shall require replacement with a long randomly generated secret.

REC-08 [Must] Recovery shall support adding a replacement email and importing a verified encrypted administrative backup, with a complete audit record and notification.

REC-09 [Must] Recovery-code verification shall be rate-limited, progressively locked after repeated failures and monitored for suspicious patterns.

12.3 Master access code and privileged actions

ADM-11 [Must] The product shall maintain a privileged master access-code mechanism for designated administrative operations, separate from Pro Mode entitlement and separate from Development Mode activation.

ADM-12 [Must] Authorized Super Admins shall authenticate with Google login and 2FA before presenting the master access code for operations that require it.

ADM-13 [Must] The Main Admin may change the master access code. The change shall require fresh Google authentication, 2FA and validation of the current code; the old code shall be invalidated immediately.

ADM-14 [Must] The master access code shall be stored as a strong verifier using a modern password-hashing scheme with unique salt and appropriate parameters. It shall never be embedded as plaintext in the APK or displayed in dashboards/logs.

ADM-15 [Must] The current bootstrap value is not repeated in this document. A low-entropy bootstrap value must be rotated to a high-entropy secret before production.

ADM-16 [Must] When a Super Admin email is replaced or supplemented during recovery, the previous email shall remain registered until an authorized administrator explicitly removes it. Replacement shall not silently revoke the former identity.

12.4 Audit and monitoring

AUDIT-01 [Must] Security and administrator audit records shall be append-only and tamper-evident. Administrators, including Super Admins, shall not be able to edit or delete audit records.

AUDIT-02 [Must] Security audit history shall be retained permanently under the product policy. Storage lifecycle shall archive records without changing their evidentiary integrity.

AUDIT-03 [Must] Audit events shall cover authentication, failed attempts, device registration/revocation, role changes, ownership transfers, code changes, recovery, exports, backup/restore, storage-provider changes, lockdown and sensitive data access.

AUDIT-04 [Must] Both Super Admins shall be able to view all administrator activity, subject to masking of secrets, passwords, tokens, private keys and raw biometric data.

AUDIT-05 [Must] Every access to audit logs shall itself be recorded.

AUDIT-06 [Must] Automated behavior monitoring shall flag unfamiliar devices/locations, unusual user-record access, repeated sensitive changes, permission changes, abnormal access patterns/times and unusual recovery or ownership operations.

AUDIT-07 [Must] Monitoring shall use risk-based thresholds and allow review to reduce false positives. It shall not silently grant privileges or automatically delete records.

12.5 Security alerts and emergency controls

SEC-01 [Must] Comprehensive security alerts shall cover unauthorized admin logins, code changes, ownership transfer, recovery, suspicious device changes, repeated failures and unusual behavior.

SEC-02 [Must] Security alerts shall use email, admin dashboard and Android push notification for both Super Admins unless an explicit exception is documented, such as new-device login notice being sent only to the signing-in Super Admin.

SEC-03 [Must] The system shall support temporary restrictions on suspicious accounts/devices, session revocation, notifications and evidence preservation.

SEC-04 [Must] Either Super Admin may independently activate global lockdown.

SEC-05 [Must] Global lockdown shall have no automatic expiry. Both Super Admins shall receive reminders every three hours, each containing a current security report.

SEC-06 [Must] The lockdown report shall summarize login attempts, blocked devices, suspicious activity, current status, administrator actions and pending requests.

SEC-07 [Must] Either Super Admin may independently deactivate global lockdown, but deactivation shall require fresh authentication, 2FA and the master access code, plus a stated reason and permanent audit entry.

SEC-08 [Must] Global lockdown activation/deactivation shall notify both Super Admins through email, dashboard and Android push.

12.6 Three-layer security model

Layer	Requirement	Engineering interpretation
Transport	TLS 1.3	Protect API/network traffic in transit. Endpoints must reject invalid certificates and use secure certificate validation.
Application payload	Authenticated payload encryption such as AES-256-GCM	Sensitive request/response payloads may be envelope-encrypted at application level with nonce, key ID, authentication tag, replay protection and authenticated identity context.
Storage at rest	Managed server-side encryption at rest	Encrypt databases, backups and object storage with managed key rotation and least-privilege access. This is not a second transport layer.

The three layers are complementary, not interchangeable. TLS protects transport; application payload encryption protects selected data beyond the transport termination point; server-side encryption protects stored data. Authentication, authorization, signatures, replay protection, audit and key lifecycle controls remain mandatory in addition to encryption.

13. Data model and information classification

13.1 Core entities

Entity	Key fields / purpose	Sensitivity
MediaItem	Stable local/Drive reference, source, path/Drive ID, title, format, duration, resolution, codec metadata, thumbnail reference.	Personal content metadata
PlaybackState	Media reference, last position, speed, selected tracks, resume preference, last played.	Personal usage data
Playlist / Favorite	Playlist ID, item references, order, title, favorite grouping.	Personal usage data
VaultObject	Encrypted object ID, nonce/tag metadata, key reference, ciphertext location, transaction state.	Highly sensitive
OfflineDownload	Drive ID, owner, destination, file metadata, expiry, extension state, signature.	Account/entitlement data
BackupManifest	Backup ID, version, type, date, content manifest, encrypted key envelope, integrity hash.	Highly sensitive
AdminAccount	Preapproved identity, role, owner flag, 2FA state, account state.	Privileged identity
AdminDevice	Registration ID, public key, device metadata, trusted status, session list.	Privileged security data
AdminSession	Session ID, issue/expiry/last activity, device ID, revocation status.	Privileged security data
AuditEvent	Immutable event ID, actor, action, target, timestamp, context, previous hash/signature.	Security evidence
CloudAccount	Provider, encrypted token reference, scopes, capacity/health, connection state.	Credential metadata

13.2 Data that must never be logged or exported in plaintext

- Recovery code and master access code
- Google OAuth refresh/access tokens and provider credentials
- Passwords, OTP/2FA codes and raw authentication challenge responses
- Private keys, raw vault keys and unwrapped backup data-encryption keys
- Raw biometric templates or raw face data
- Plaintext Secret Vault media names/content where the vault metadata model requires encryption

13.3 Retention principles

Playback history is retained until the user clears it. Offline Drive copies follow the 30-day validity plus five-day grace and approved-extension process. Cloud-offloaded backups are retained indefinitely until authorized

deletion. Security audit logs are permanent and immutable. Other data categories shall receive explicit retention periods during implementation and privacy review.

14. External interfaces and integration requirements

14.1 Android interfaces

- Media session and notification controls
- Foreground service for eligible background audio/download tasks
- Storage Access Framework and scoped storage APIs
- Picture-in-Picture and multi-window APIs
- Android Keystore and biometric prompt
- Audio routing/device callback APIs
- Display mode and refresh-rate APIs where supported
- Network/battery/thermal signals within OS permissions and policy
- APK update/download and package installer handoff

14.2 Google identity and Drive

Google Sign-In identity and Google Drive file scopes shall be requested separately and with least privilege. The app shall support account linking/disconnection, token revocation, safe refresh and clear UI for the difference between app identity and Drive access.

14.3 Subtitle and lyrics providers

Provider adapters shall isolate external APIs, rate limits, attribution requirements, content matching, caching and failure states. User consent and provider terms must be respected. Local subtitle functionality shall remain available when providers are offline.

14.4 Backup cloud providers

Each storage provider shall be implemented behind a capability interface. The adapter shall report whether it supports list, upload, download, checksum/verification, deletion, quota information and account disconnection. Unsupported operations shall be disabled or handled with an explicit warning.

14.5 Backend API groups

API group	Representative operations
Identity and devices	Admin login callbacks, 2FA verification state, device registration, session listing/revocation, device-key challenge.
Development Mode	Code validation, activation, expiry, device entitlement, revocation, audit events.
Offline download policy	Register download metadata, expiry status, extension request, approval/rejection, signed policy retrieval.
Administrator governance	Role changes, ownership transfer, master/recovery configuration, global lockdown.
Backup management	Backup manifest registration, storage providers, replication jobs, restore points, health reports and restore approvals.
Notifications	Email, dashboard, Android push events and alert preferences.

15. Non-functional requirements

15.1 Performance and responsiveness

NFR-P01 **[Must]** Core UI navigation and controls shall remain responsive while media scan, conversion, backup or download tasks run in the background.

NFR-P02 **[Must]** The application shall avoid blocking the UI thread with decoding, database scans, file hashing, encryption or conversion work.

NFR-P03 **[Must]** Playback startup, seek response and rendering performance shall be benchmarked across low-, mid- and high-tier Android devices. Target values shall be set after prototype profiling on representative hardware.

NFR-P04 **[Must]** The library shall use paging/virtualization and incremental indexing so large libraries do not require loading all records and thumbnails into memory at once.

NFR-P05 **[Must]** Thumbnail and media metadata caching shall be bounded and managed automatically.

NFR-P06 **[Must]** Processing and parallel jobs shall adapt to memory, battery, thermal and hardware constraints. The user may override only within safe bounds.

15.2 Reliability and recovery

NFR-R01 **[Must]** Core local playback shall remain available without network access.

NFR-R02 **[Must]** File operations that move, vault, repair, convert or replace files shall be transactional or use temporary outputs so partial failure does not destroy the source.

NFR-R03 **[Must]** Backup, restore, sync and download jobs shall persist recoverable state and support retry after application restart where technically possible.

NFR-R04 **[Must]** A failed restore shall return the system to its pre-restore state or preserve an explicitly recoverable snapshot.

NFR-R05 **[Must]** Application crashes shall not silently discard unfinished job metadata or mark incomplete encrypted files as valid.

15.3 Security

NFR-S01 **[Must]** Secrets shall be stored using platform/managed secure storage and shall never be hardcoded in APK resources, source-controlled configuration or logs.

NFR-S02 **[Must]** Use least-privilege permissions, short-lived access tokens, refresh-token protection, role checks and server-authoritative decisions.

NFR-S03 **[Must]** Cryptographic operations shall use vetted libraries and safe nonce/IV management. Custom cryptography is prohibited.

NFR-S04 **[Must]** All security-sensitive endpoints shall validate authorization on the server, not only in the client UI.

NFR-S05 **[Must]** Sensitive actions shall have audit records and appropriate reauthentication requirements.

15.4 Privacy and accessibility

NFR-PR01 **[Must]** Analytics/usage collection shall be consent-gated during first installation and remain disabled when consent is refused.

NFR-PR02 **[Must]** Crash/diagnostic reporting shall be consent-gated per report unless the user explicitly enables future consent.

NFR-PR03 **[Must]** The product shall provide app-specific accessibility controls and follow Android font scaling, screen reader and system accessibility preferences.

NFR-PR04 **[Must]** Sensitive Vault surfaces shall be protected from screenshots/recording and Recent Apps previews where supported.

15.5 Maintainability and observability

NFR-M01 **[Must]** Media engine, decoder, renderer, processing stages, cloud providers and backup providers shall be separately replaceable modules.

NFR-M02 **[Must]** Diagnostics shall include non-sensitive technical state such as engine path, decoder choice, codec metadata and failure code, but must redact credentials and content.

NFR-M03 **[Must]** Builds shall support reproducible configuration, automated tests and staged rollout of native processing changes.

NFR-M04 **[Must]** The system shall maintain compatibility/version fields for backups, encrypted manifests, settings schemas and native plugin interfaces.

16. Key workflows and state transitions

16.1 Playback and decoder fallback

1. User selects a local or Google Drive media item.
2. The source adapter validates access and retrieves metadata/stream information.
3. The capability manager probes available decoders, rendering paths, HDR support and output route.
4. The policy selects HW, HW+, SW or hybrid path based on compatibility and prior failure signals.
5. Playback starts and emits status/diagnostic events.
6. If decoding or rendering fails, the recovery controller retries an alternative compatible path.
7. If damage remains, attempt frame recovery and skip irrecoverable frames where safe.
8. If all paths fail, show a clear error with optional consent-based report submission.

16.2 Vault move transaction

1. Authorize source read/write access.
2. Create transaction ID and encrypted staging output.
3. Stream encryption and calculate authenticated integrity state.
4. Verify staged ciphertext and metadata.
5. Atomically commit canonical Vault object.
6. Only then delete original when the user explicitly selected move.
7. On error/crash, preserve source and safely clean or resume staging.

16.3 Offline Drive expiry

1. User downloads the original-quality file and selects destination.
2. Backend records stable Drive ID, owner, metadata, original timestamp and initial 30-day expiry.
3. Client stores signed expiry metadata and schedules daily in-app and OS notices.
4. Within the five-day grace period, the user may request an extension by designated email.
5. Admin approves one extension of 1-30 days for all offline downloads of that user, or rejects the request.
6. If rejected, the user is notified immediately and may submit a new request while grace remains.

7. If no approval is valid at deadline, local offline copy is deleted at the next enforceable opportunity; original Drive file is untouched.
8. Next monthly approval cycle is scheduled from the resulting policy timestamp.

16.4 Backup offload

1. Primary backup storage usage is monitored.
2. At 70% usage, select oldest eligible backups and calculate the minimum amount to offload.
3. Choose the currently designated secondary account and test its health/capabilities.
4. Compress if not already compressed, encrypt locally, upload and verify integrity/retrieval.
5. Only after successful verification, remove primary archive while retaining manifest and restore-point reference.
6. If the account is disconnected/full/unavailable, pause and ask the Main Admin whether to use another account or stop.
7. If primary storage reaches the 5 GB cap without safe offload capacity, request Main Admin decision to replace eligible older backups or add storage.

16.5 Administrator recovery

1. Super Admin begins recovery from an available device or recovery interface.
2. The service challenges the registered device key and verifies cryptographic device ownership and historical registration.
3. If device proof succeeds, restore the authorized account access path after risk checks.
4. If device verification is unavailable or fails, request the recovery/reset code.
5. A valid recovery code permits controlled access reset, email replacement/addition and encrypted administrator-backup import.
6. Record all attempts, verification results and resulting changes in immutable audit history.
7. Notify the appropriate administrators according to the recovery-notification policy.

16.6 Global lockdown

1. Either Super Admin initiates lockdown and reauthenticates as required.
2. Backend marks protected operations and configured access paths as locked down.
3. Sessions/tokens may be restricted or revoked according to incident policy.
4. Both Super Admins receive email, dashboard and push alerts.
5. Every three hours, both receive a current security report while lockdown remains active.
6. Either Super Admin may deactivate after fresh auth, 2FA, master code and a recorded reason.

17. Acceptance criteria

The following criteria are release-gate examples. Detailed device matrices, quantitative performance targets and exact provider API contracts shall be baselined during implementation planning.

Area	Acceptance condition
Responsive UI	Video, Audio, Tools and Settings adapt across tested phone/tablet/window sizes and orientations without clipped core controls.

Area	Acceptance condition
Decoder fallback	Inject or simulate a decoder initialization failure and verify automatic fallback through an alternative supported HW/HW+/SW/hybrid path.
Gestures	A single tap reveals controls; controls hide; left/right double-tap seeks 10 seconds; center Previous/Next changes media items.
HDR	HDR content uses supported HDR output; on unsupported SDR path, tone mapping produces visible output without a crash.
Audio	Background audio follows Settings; headset controls work where platform route exposes them; queue and playlist import/export pass tests.
Library	All accessible supported media is indexed; exclusions and hidden-file settings are respected; duplicate detection never silently deletes source media.
Subtitles	Local subtitle auto-match and manual selection work; offset can be adjusted; unsupported format fails gracefully.
Tools	Conversion preserves the source according to chosen preference; batch queue can pause/resume/cancel/reorder; audio extraction preserves original stream when possible.
Enhancement	Deblur, denoise and upscale stages can be individually enabled/disabled and tested; output is labelled as upscaled where appropriate.
Drive downloads	Original bytes are preserved; pause/resume/retry work; expiry warnings appear; expiry deletes only offline copy and never original Drive item.
Vault	Interrupted move never deletes original before verified encrypted commit; Vault data is not exposed to ordinary media indexing.
Backups	Archive is compressed before encryption, has independent key and integrity ID, verifies after creation and restores in isolated tests.
Offload	Primary backup is removed only after remote retrieval verification; failed cloud account causes Main Admin prompt and no silent provider switch.
Admin security	Admin login requires preapproved Google identity and 2FA; 30-minute idle timeout; max two active devices per Super Admin.
Audit	Administrator actions are append-only, tamper-evident and cannot be edited/deleted through admin UI/API.
Recovery	Device-key verification and recovery-code fallback work in controlled tests; recovery code is never logged or stored in plaintext.
Consent	Analytics and crash reporting remain disabled without user consent and respect later Settings changes.

18. Verification and test strategy

18.1 Test levels

- Unit tests for scanner rules, decoder policy, subtitle matching, expiry calculations, role/permission checks, backup manifests, encryption wrappers and processing stages.
- Integration tests for Flutter/native boundary, engine adapter, Android storage access, Keystore, Drive API, notification services, provider adapters and backend APIs.
- End-to-end tests for playback, download, conversion, Vault movement, backup, restoration, administrator login/recovery and lockdown.
- Device compatibility tests across low/mid/high tiers, multiple Android versions, SoC families, hardware decoders, Vulkan support, HDR display paths, audio output paths and thermal conditions.
- Security tests including threat modeling, permission bypass attempts, replay attempts, token leakage checks, tamper detection, rate limiting, session revocation and storage inspection.
- Resilience tests including process death, reboot, network loss, provider outage, corrupted media, partial upload, disk full, backup corruption and interrupted restore.
- Accessibility tests for screen reader navigation, focus order, large text, contrast, reduced motion and touch target sizes.

18.2 Mandatory failure-injection scenarios

- Kill the process during Vault staging, commit and original deletion phases.
- Interrupt a cloud upload at several offsets and verify that incomplete objects are not considered valid replicas.
- Corrupt a backup manifest, ciphertext block or authentication tag and verify rejection plus previous-valid-backup fallback.
- Expire a Drive offline copy while the device is offline, then reconnect and verify signed expiry enforcement and original-file preservation.
- Attempt a third administrator device session and verify explicit choice of session to terminate.
- Revoke an administrator device and verify immediate session/token invalidation.
- Attempt admin audit modification/deletion through UI and API; verify denial and audit of the attempt.
- Force a decoder failure and verify alternate decoder path without losing playback position unnecessarily.
- Run processing under low battery/high thermal state and verify adaptive reduction/override behavior.

18.3 Security review gates

TEST-S01 **[Must]** No release may ship with hardcoded recovery, master, development or provider credentials.

TEST-S02 **[Must]** No release may ship with unresolved critical/high severity security findings in authentication, Vault encryption, backup encryption, admin authorization or cloud token handling.

TEST-S03 **[Must]** The security team shall review cryptographic key lifecycle, recovery, rotation and compromise scenarios before production activation.

19. Delivery phases and release gates

Phase	Primary scope	Exit gate
0. Technical validation	Engine/library licensing; Android device capability survey; Flutter/native bridge spike; cloud provider API feasibility; security architecture review.	Approved component shortlist, risk register, architecture decisions and proof-of-concept playback.

Phase	Primary scope	Exit gate
1. Core player foundation	Flutter shell, Video/Audio/Tools/Settings navigation, local scanning, core decoder paths, playback controls, preferences and crash UI.	Core local playback, responsive UI, library scan and fallback tests pass.
2. Player completeness	Subtitles, audio player, history/resume, playlists, PiP, gestures, HDR/refresh matching and initial tools.	Functional acceptance for video/audio/subtitle workflows.
3. Pro and enhancement	Pro UI, profiles, graphics detection, deblur/denoise/upscale pipeline, preview and conversion pipeline.	Device benchmarks and independent processing-stage test evidence.
4. Cloud and offline	Google Drive browsing/streaming, download manager, original-quality copies, expiry/extension policy and Drive sync.	Drive OAuth/security review and offline expiry tests.
5. Vault and backup	Transactional Vault, key recovery, app backups, restore points, cloud provider adapters and offload.	Crypto review, interrupted-operation tests and successful isolated restore tests.
6. Admin and Dev Mode	Admin login/2FA, role system, device sessions, audit/monitoring, Dev Mode activation, recovery and lockdown.	Penetration testing, permission matrix validation and recovery/lockdown drills.
7. Stabilization and release	Accessibility, localization, performance, thermal testing, APK update flow, documentation and support runbooks.	Release checklist, privacy consent validation and signed release package.

Development sequencing may overlap, but cryptographic foundations, data schemas, engine selection and provider feasibility should be validated early because they influence many downstream components.

20. Risks, dependencies and mitigations

Risk / dependency	Impact	Mitigation / decision
Device fragmentation	Different SoCs and Android versions expose different decoder, GPU, NPU and HDR capabilities.	Capability-probe at runtime; maintain fallback paths; validate device tiers; avoid hard-coded assumptions.
Upscaling expectations	Users may interpret 360p-to-1080p output as native 1080p quality.	Label upscaled output; provide before/after preview; document that new detail is estimated, not recovered certainty.
Thermal/battery limits	Heavy processing can heat devices and degrade battery/performance.	Adaptive pipeline, thermal-aware concurrency, Eco/Balanced/Performance modes and safe manual overrides.
Cloud provider API coverage	Some providers may not offer suitable public APIs for all selected operations.	Prototype provider capability early; only support official/approved APIs; expose capability matrix and partial support.
Offline expiry enforcement	Remote commands cannot reach disconnected/powered-off devices.	Signed local expiry records, secure timestamps where available, enforcement at next app execution and server-side denial on reconnect; state residual limitation.

Risk / dependency	Impact	Mitigation / decision
Device identity	Android IDs and fingerprints are mutable/spoofable.	Bind using registered device keys and hardware-backed Keystore where available; treat identifiers as supplemental signals.
Vault recovery escrow	Server-side escrow creates a high-value recovery target.	Encrypt escrow, separate duties, harden key access, alert on recovery, audit every operation and conduct external security review.
Backup key loss	If all recovery mechanisms are lost, encrypted backups may be unrecoverable.	Multiple carefully separated recovery paths, offline recovery material, verified key rotation and periodic restore drills.
Crypto misuse	Nonce reuse, weak KDF parameters or secret leakage can invalidate encryption.	Use vetted libraries, unique nonces, per-backup keys, key wrapping, code review and cryptographic tests.
Third-party licensing	Media codecs and engines may impose redistribution/attribution requirements.	Complete legal/licensing review before selecting and packaging libraries.
Storage exhaustion	Primary backup quota or device storage may fill.	70% offload trigger, 5 GB quota warning, admin prompts, no silent deletion and storage dashboard.
Administrative compromise	Compromised Super Admin credentials can threaten the whole system.	Mandatory 2FA, device verification, session limits, progressive lockout, immutable logs, alerts and lockdown.

21. Open implementation decisions

The product behavior is baselined, but the following engineering choices must be completed during architecture and sprint planning. They are not permission to silently change the agreed behavior.

- Final media engine and codec-library selection, including license, ABI, maintenance and device-compatibility assessment.
- Exact Android minimum/target SDK policy and supported device/version matrix.
- Concrete backend hosting, database, queue, object storage, key management service and observability stack.
- Exact Google Drive scopes and OAuth consent model, plus token revocation and disconnected-account behavior.
- Provider-by-provider feasibility for OneDrive, Dropbox and multiple Proton Drive accounts; supported operation matrix and rate limits.
- Exact backup schedule times, time zone behavior, full-backup interval implementation and whether critical-data snapshots are separate from general incremental snapshots.
- Precise monthly extension renewal rule and how the next 30-day validity period interacts with an Admin-approved 1-30 day extension.
- How offline expiry handles user-cleared app data, APK uninstall/reinstall, device clock rollback and long periods without network.
- Exact recovery code entropy, provisioning and key-wrapping design. The initial bootstrap value must be rotated before production.
- Whether the privileged master access code and recovery/reset code use distinct secrets in production. They should be distinct even if an initial bootstrap value was informally reused during discussion.

- Exact hardware support tiers and minimum performance targets for 1080p/4K, high frame rates, HDR and real-time enhancement.
- Which enhancement models/scalers are approved, their license terms, storage footprint, quality metrics and supported ABI/GPU/NPU paths.
- Exact external subtitle and lyrics providers, legal terms, attribution, rate limits and cache policy.
- Default inactivity timeout for the Secret Vault beyond the selected on-background lock behavior.
- Quantitative thresholds for 70% backup offload, primary 5 GB limit and backup-health alert severities.
- Operational ownership of the designated project email, manual Development Mode review queue and admin on-call/escalation procedures.

Baseline rule

If an implementation detail is not specified, engineering should propose the safest maintainable default, record it as an Architecture Decision Record (ADR), and seek product-owner approval when the choice affects user-visible behavior, privacy, data retention, security or data loss.

Appendix A. Consolidated decision traceability

This appendix maps the question batches from requirements discovery to the corresponding SRS areas. Individual question identifiers are retained so later changes can be traced.

Decision range	Topics captured	SRS locations
Q77-Q95	Admin inactivity session, concurrent devices, new-device sign-in, device names/revocation, failed-login lockout, recovery, recovery code.	Sections 3, 12, 13, 16
Q96-Q115	Backup schedule, 5 GB target, full/incremental backup, restore points, integrity, corruption recovery, admin permissions, provider support and key security.	Sections 11, 13-18
Q116-Q135	Multiple storage accounts, 70% offload threshold, oldest-first offload, verified copy removal, provider failure prompts, compression/encryption, selective restore and recovery.	Sections 11, 14, 16-18
Q136-Q145	Weekly full backups, event/daily increments, retries, encrypted ZIP archival, health reports and restore mode.	Sections 11, 16-18
Q146-Q165	Decoder fallback, damaged frames, HDR, refresh matching, passthrough, subtitle format and sync, PiP, orientation and gestures.	Sections 7, 15, 17-18
Q166-Q175	Background audio, Bluetooth controls, queue, EQ, volume safety, profiles, lyrics, sleep timer and playlist formats.	Sections 7, 17-18
Q176-Q185	Separate libraries, full scanning, exclusions, hidden files, duplicates, layouts, resolution filters and sorting.	Sections 6-7

Decision range	Topics captured	SRS locations
Q186-Q195	Conversion, quality targets, multi-stage upscaling, source preservation, queues, extraction, frame tools, subtitles and media details.	Sections 7-8
Q196-Q205	Automatic enhancement, separate pipeline stages, deblur/denoise, presets, thermal adaptation, thumbnails, repair and comparison.	Sections 8, 17-18
Q206-Q215	Playback speed, frame stepping, A-B loop, bookmarks, resume, power modes, profiles, per-video preferences, themes and Pro customization.	Sections 6-8
Q216-Q225	Language, opening screen, search, favorites, history, deletion confirmation, manual APK updates, consent, notifications and accessibility.	Sections 6-7, 15
Q226-Q235	PIN/biometric app lock, Vault locking, screenshot prevention, recents privacy, permissions, storage access, background downloads, external displays and data clearing.	Sections 7, 10, 14-18
Earlier confirmed baseline	Flutter/Dart + C/C++, local + Google Drive media sources, Pro and Dev separation, Vault, Dev Mode lifecycle, Drive expiry and admin ownership/security decisions.	Sections 1-5, 8-14, 20-21

Appendix B. Glossary and abbreviations

Term	Definition
A-B repeat	Loop playback between a user-selected start point (A) and end point (B).
AES-256-GCM	Authenticated encryption with 256-bit AES keys and Galois/Counter Mode; requires safe unique nonce management.
Admin / Support Admin	Delegated roles with narrower permissions than Super Admin.
Backup DEK	Data-encryption key generated for an individual backup archive.
Device binding	Association of an account/entitlement with registered cryptographic device identity, not merely a display name or Android ID.
Development Mode	Online-validated, time-limited, device-bound mode for experimental processing and hardware overrides.
HW / HW+ / SW	Hardware decoding paths, enhanced/hybrid hardware paths as exposed by the selected engine, and software decoding. Exact semantics must be documented for the chosen engine.

Term	Definition
Incremental backup	Backup containing data changed since a prior full or incremental backup, according to the selected chain strategy.
Main Admin / protected owner	The designated protected-owner Super Admin with the highest ownership/governance responsibilities.
NPU	Neural Processing Unit; availability and API access depend on device and vendor implementation.
PiP	Picture-in-Picture playback window supported by Android.
Pro Mode	Freely available advanced user-facing controls; not a security or entitlement gate.
Restore point	Verified recovery checkpoint created before important operations or according to backup policy.
SRS	Software Requirements Specification.
TLS 1.3	Transport Layer Security protocol version 1.3.
Vault	Encrypted private media area with protected metadata, access controls and transactional file movement.
Vulkan	Cross-platform graphics/compute API; availability is detected at runtime and cannot be assumed on every device.

Baseline sign-off statement

This document consolidates the product decisions captured through the requirements-discovery process into a single implementation-oriented SRS. It is suitable as the starting point for UX wireframes, technical architecture, backlog decomposition, interface contracts and QA planning. Before production, the open decisions in Section 21 must be resolved and the security-sensitive designs must pass specialist review.

Release-critical reminders

Do not hardcode the bootstrap recovery/master secrets. Rotate weak bootstrap credentials before production. Do not send encrypted backup contents as plaintext email attachments. Do not silently delete backups, offline originals or source media. Do not enable Development Mode offline. Do not claim hardware, HDR, Vulkan, NPU, bit-perfect or cloud-provider capabilities without runtime/provider verification.